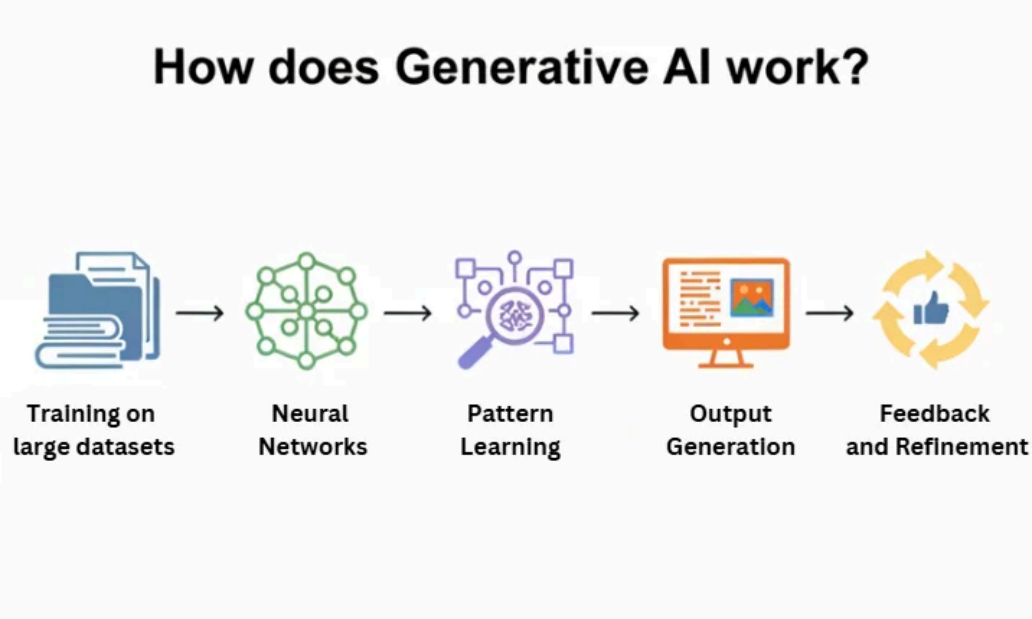
GENERATIVE AI

TABLE OF INDEX

Index		Title	Pages
1		What is Gen AI?	3
2		Types of Generative AI by Output	4
	A	Text Generation and Summarization	6
	B	Image and Video Generation	7
	C	Synthetic Voice and Music Creation	8
	D	Code Generation and Automation	9

1. What is Generative AI?

Generative artificial intelligence (**GenAI**) is a subfield of artificial intelligence (AI) that uses generative models to generate text, images, videos, audio, computer code or other forms of digital data.These models learn the underlying patterns and structures of their training data, and use them to generate new data in response to input, which often takes the form of natural language prompts.



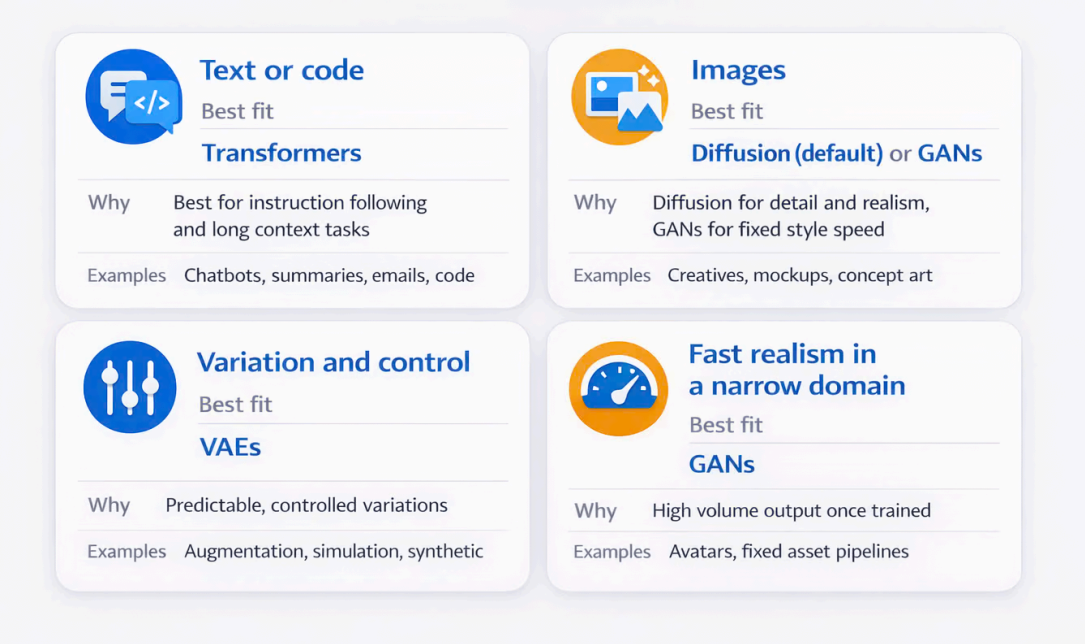
Generative AI learns patterns from large amounts of training data and uses those patterns to generate new content. In simple words, Generative AI is technology that can generate something new when you give it instructions.

Example: If you ask an AI to write a story about a robot, it uses patterns learned during training to generate a story rather than simply copying one complete story from its training data.

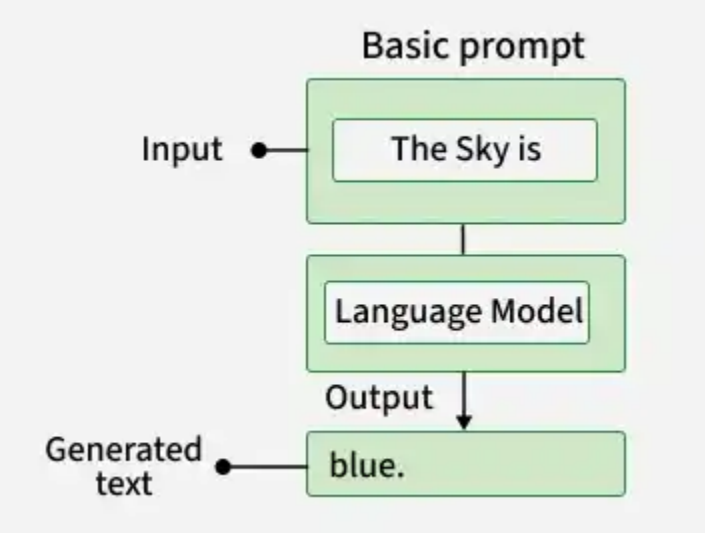
2. Types of Generative AI by Output

Output type	How it works	What it generates
Text generators	Learns patterns in large text datasets and predicts the next words to produce coherent text	Emails, summaries, answers, scripts, long-form content
Image generators	Learns visual patterns from large image datasets and maps text prompts to images	Visuals from prompts, concept art, marketing creatives, design variations
Audio and voice generators	Learns pronunciation, tone, and rhythm from audio data to produce speech from text or a voice sample	Speech, voiceovers, synthetic voices, spoken audio
Video generators	Generates a sequence of frames while trying to keep subject and	Short clips, animations, prompt-based scenes

	scene continuity	
Code generators	Learns syntax and programming patterns from code corpora to suggest or generate working code	Code snippets, functions, tests, refactors



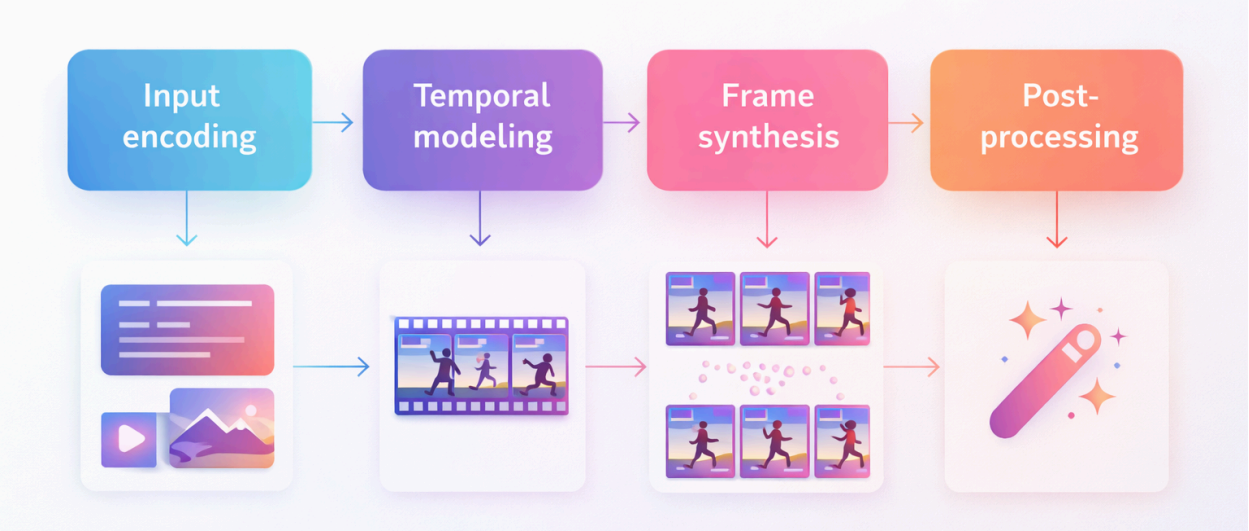
A. Text Generation and Summarization: Transformer-based models like GPT and BERT power AI-driven chatbots, automated content creation, and summarization tools used in customer service, publishing, and enterprise knowledge management.



Use Cases:

- **Knowledge Retrieval & Enterprise Search:** Grounding responses in private company documentation or data repositories (such as RAG systems) to answer internal employee or customer queries.
- **Conversational AI & Chatbots:** Powers virtual customer service help desks, interactive assistants, and automated support desks using decoder models like GPT.
- **Text Summarization:** Condenses long research papers, legal documents, meeting transcripts, and books into concise, readable summaries.
- **Machine Translation:** Translates text fluidly between multiple global languages by mapping contextual tokens across sequences.

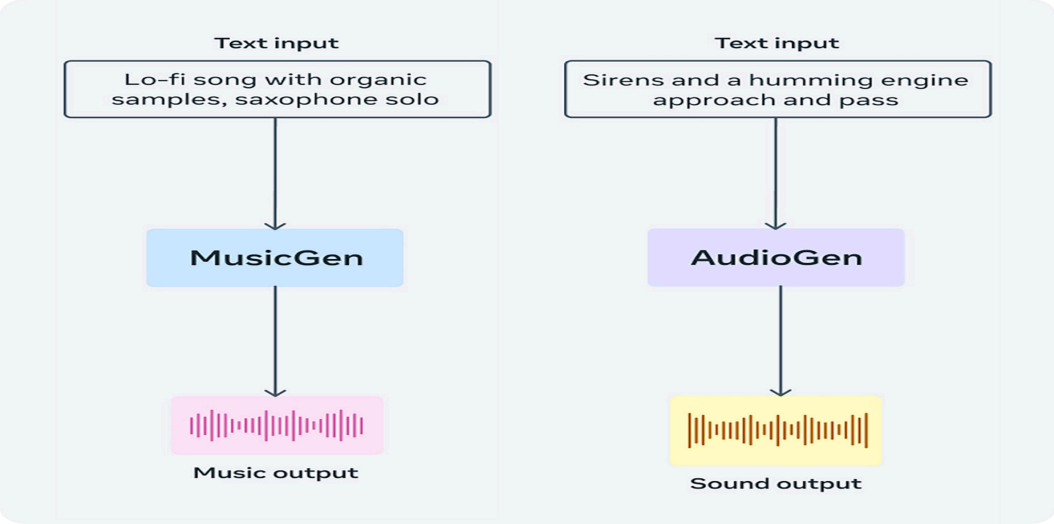
B. Image and Video Generation: GANs and diffusion models are widely used in media, fashion, and gaming to create hyper-realistic visuals, synthetic training data, and even virtual try-on experiences.



Use Case:

- **Product & E-Commerce Marketing:** Automatically place products into seasonal or lifestyle backgrounds to scale marketing assets across platforms like Instagram and Facebook.
- **Image Editing & Inpainting:** Add, remove, or modify specific elements within a photo without manual design software.
- **Image-to-Video Animation:** Turn static photos, illustrations, or digital art into short dynamic clips or B-roll using tools like Leiapix. [1]
- **Missing B-Roll Generation:** Synthesize realistic missing clips or background scenes when physical re-shoots are impossible.

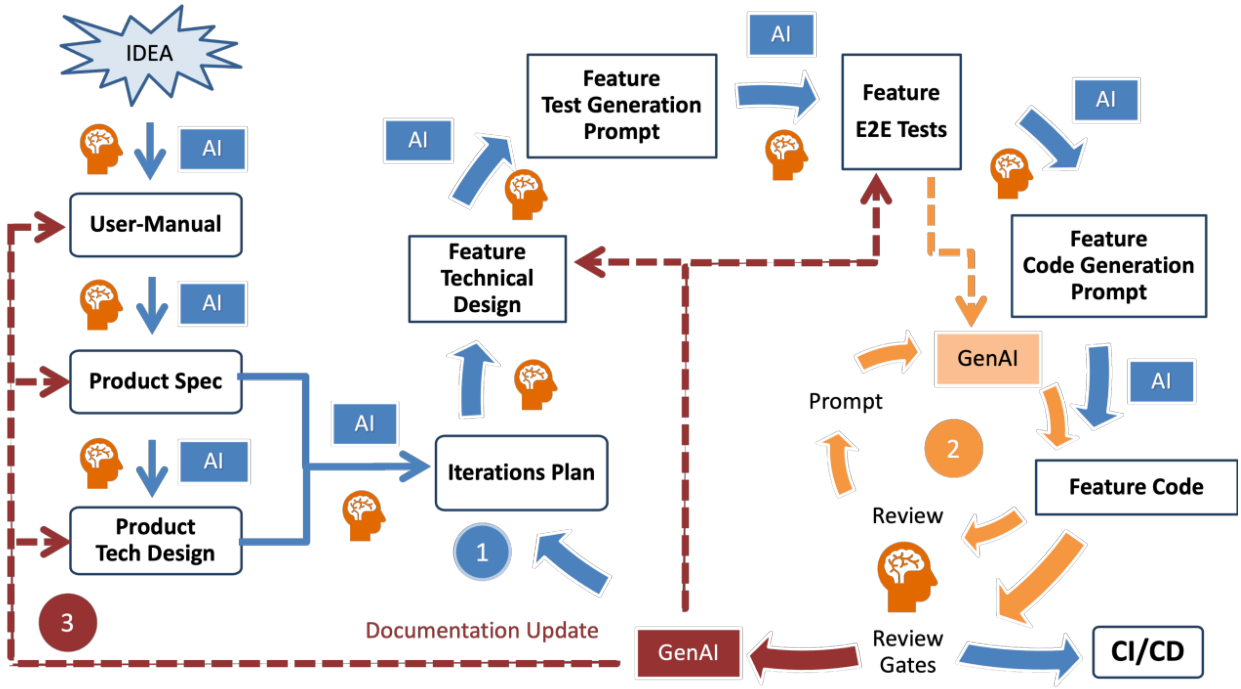
C. Synthetic Voice and Music Creation: Models trained on audio data can generate lifelike speech for virtual assistants, audiobooks, or voiceovers, and compose original music tracks for entertainment or branding.



Use Case:

- **Text-to-Speech (TTS) & Voice Cloning:** Convert written text into natural-sounding voiceovers for audiobooks, video narration, and virtual assistants, or clone a specific voice using short audio samples (such as Gemini TTS or ElevenLabs).
- **Speech-to-Speech (STS) & Audio Localization:** Translate spoken content into different languages while preserving the original speaker's timbre and emotional delivery, as seen in localized news broadcasts and podcasts.
- **Music Composition & Production:** Generate original background tracks, melodies, harmonies, and full musical arrangements based on style or text descriptions for ads, games, and films.
- **Sound Effect & Ambient Audio Design:** Synthesize custom Foley, object interactions, and environmental background soundscapes for video games, movies, and virtual worlds

D. Code Generation and Automation: Codex-style models are being used to write code snippets, automate repetitive programming tasks, and support developers in integrated development environments.



Use Case:

- **Boilerplate & Script Automation:** Instantly generates standard template setups, configuration files, and routine code chunks.
- **Intelligent Code Completion:** Context-aware code filling within the IDE as the developer types.
- **Legacy Code Refactoring:** Modernizes older codebases, optimizes algorithms, and translates logic across languages (e.g., COBOL to Java).
- **Automated Testing & QA:** Writes comprehensive unit tests, boundary test cases, and edge-case validations.